

Table S12. Visual light sensitivity

Metric	Scale	Association per VLSQ-8 SD (95% CI)	FDR-adjusted p	Participants / participant-days / sites	
Mean melEDI	Ratio (percentage change)	×0.915 (0.811–1.031)	0.221	141 / 816 / 9	
Brightest 10 h mean	Ratio (percentage change)	×0.851 (0.716–1.013)	0.160	141 / 816 / 9	
Darkest 10 h mean	Ratio (percentage change)	×1.024 (0.934–1.123)	0.689	141 / 816 / 9	
Time above 1,000 lx melEDI	Ratio (percentage change)	×0.876 (0.765–1.002)	0.160	141 / 816 / 9	
Time above 250 lx melEDI during wake	Ratio (percentage change)	×0.900 (0.806–1.005)	0.160	141 / 737 / 9	
Time below 10 lx melEDI before sleep	Difference in hours	+0.063 h (-0.062–+0.188 h)	0.422	139 / 655 / 9	
Time below 1 lx melEDI during sleep	Ratio (percentage change)	×0.998 (0.964–1.033)	0.903	141 / 778 / 9	
Longest continuous period above 250 lx melEDI	Ratio (percentage change)	×0.917 (0.828–1.014)	0.167	141 / 816 / 9	
melEDI dose	Ratio (percentage change)	×0.846 (0.716–0.999)	0.160	141 / 761 / 9	

Associations compare scores separated by one participant-level SD (5.540 VLSQ-8 points). The hour-scale row is an absolute difference; ratio rows can be read as percentage changes. Intervals are two-sided 95% Wald CIs. FDR adjustment is across all nine metrics; adjusted p-values would be bold at 0.050.